

Medical Robotics

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Final projects



SAPIENZA
UNIVERSITÀ DI ROMA

- 21+ projects on the 3 main topics: robotics, AI, biomedical
- groups of 3 or 4 students
- programming in C/C++, Matlab or Python required
- projects must be chosen by June 5th
- send me an e-mail with subject [MR: project choice], specifying at least 3 projects in a list with decreasing order of preference and the names of the students in the group
- projects are assigned only to students who have passed the midterm
- projects replace the standard exam (written + oral test)
- when ready to start working, send me an e-mail to arrange a kick-off meeting

presentation of results

- written report with selected results + pdf, software, videos, datasets, presentation (all **shared in a Drive folder**)
- **15 min** live presentation (ppt or other) of the obtained results, including Q/A
- **NOTE:** this year, after the project presentation, each component of the group will present his/her individual homework in **5 minutes**

Project I: Learning-based registration



Goal of the projects

- keep the same hardware but remove the dependence on known points and on the prescribed touching order
- the operator acquires an unordered, partial point set from the exposed phantom surface, and a learning-based method estimates the same Polaris→phantom rigid transformation by aligning that point set to the digital twin
- the existing **four-point method is the baseline** that every variant must be compared against, using the same target registration error (TRE)
- two acquisition modalities: **sweep**, **sparse** to be tested for all variants

Project 1.1: correspondence-free registration

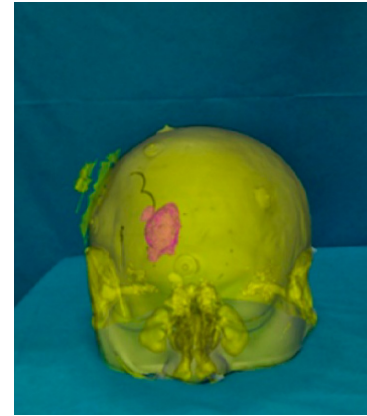
- **goal:** recover the rigid transform with no point matches
- implement the algorithm in: Y. Aoki, H. Goforth, R. A. Srivatsan, S. Lucey, “PointNetLK: Robust & Efficient Point Cloud Registration Using PointNet,” CVPR, 2019. arXiv:1903.05711. Code: <https://github.com/hmgoforth/PointNetLK>
- compare the sweep and sparse modalities
- compare against the baseline
- best suitable for students who already have, or wish to develop, a solid background in machine learning
- you can propose an alternative algorithm but the goal should remain the same

Project 1.2: correspondence-learned registration

- **goal:** learn point correspondences
- implement the algorithm in: Y. Wang, J. M. Solomon, “Deep Closest Point: Learning Representations for Point Cloud Registration,” ICCV, 2019. arXiv:1905.03304. Code: <https://github.com/WangYueFt/dcp>
- compare the sweep and sparse modalities
- compare against the baseline
- best suitable for students who already have, or wish to develop, a solid background in machine learning
- you can propose an alternative algorithm but the goal should remain the same

Project 1.3: Learned descriptors + RANSAC + ICP

- **goal:** dense 3D feature descriptors

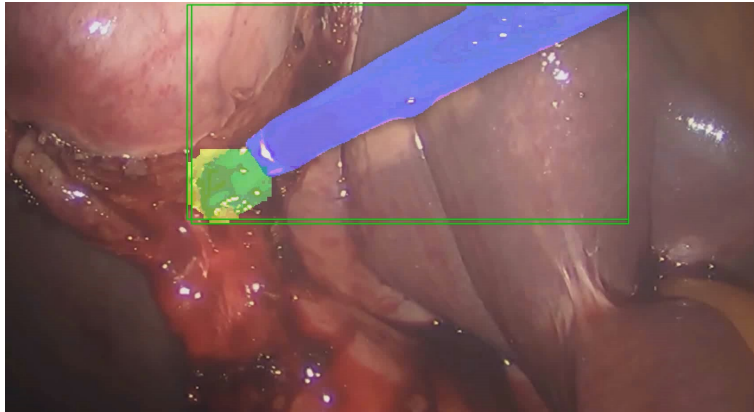


- implement the algorithm in: C. Choy, J. Park, V. Koltun, “Fully Convolutional Geometric Features,” ICCV, 2019. Code: <https://github.com/chrischoy/FCGF>
- compare the sweep and sparse modalities and against the baseline
- best suitable for students who already have, or wish to develop, a solid background in machine learning
- you can propose an alternative algorithm but the goal should remain the same

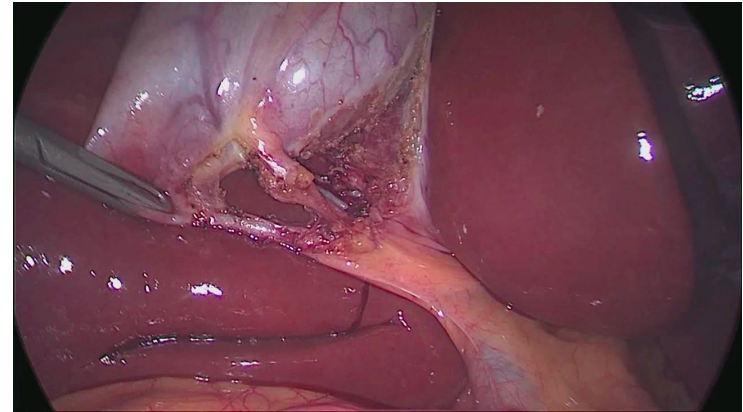
Project 4: multi-RCM for the ROSA robot

- in the [ROSA system](#) for neurosurgery the RCM is located at the skull entry point
- in some phases of the procedure it would be useful to have the RCM located at the target point and allow small conical motion at the entry point
- **goal:** allow a double RCM
- implementation: (a) 3D and kinematic model of ROSA in Unity; (b) double RCM through kinematic control
- best suitable for students who already have, or wish to develop, a solid background in robot kinematic control

Project 5: automatic surgical scene understanding



contact detection



CVS

goal of the projects

- workflow analysis, anatomy segmentation, tool detection & tracking, CVS assessment, body pose estimation, Surgical action recognition
- all projects build on the public datasets of the CAMMA research group at the University of Strasbourg
- each project has a specific task and a reference dataset

Project 5.1: surgical workflow analysis

- **goal:** develop a model that automatically recognises the current surgical phase from the video stream
- **dataset:** Cholec80 (baseline)
- **your task:** temporal modelling of video sequences; handling of class imbalance; workflow-specific metrics
- best suitable for students who already have, or wish to develop, a solid background in machine learning

Project 5.2: anatomy segmentation

- **goal:** semantic / instance segmentation of anatomical structures and instruments in endoscopic frames, with the difficulties typical of the domain (reflections, smoke, blood, occlusions)
- **dataset:** Endoscapes
- **your task:** determine the segmentation architectures, data augmentation specific to surgical imagery, qualitative error analysis
- best suitable for students who already have, or wish to develop, a solid background in machine learning

Project 5.3: tool detection & tracking

- **goal:** localise the surgical instruments and follow their trajectory over time, handling entries/exits from the field, occlusions and the simultaneous presence of several instruments
- **dataset:** CholecTrack20 (tracking) or CholecT50 / M2CAI16-tool (detection)
- **your task:** object detection, temporal association and tracking algorithms (multiobject tracking)
- best suitable for students who already have, or wish to develop, a solid background in machine learning

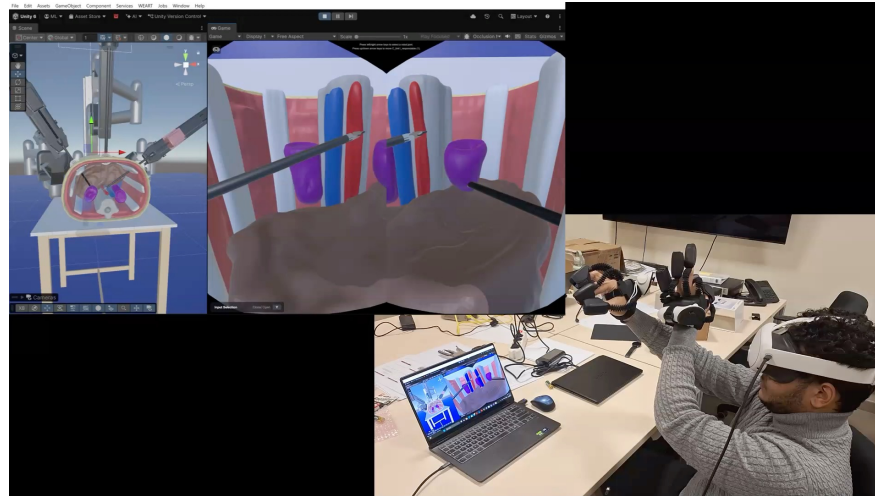
Project 5.4: Critical View of Safety (CVS) Assessment

- **goal:** automatically assess whether and to what extent the three criteria of the CVS are satisfied in the video
- **dataset:** Endoscapes
- **your task:** linking visual perception to clinical safety notions; handling of imbalanced and subjective labels; model interpretability
- best suitable for students who already have, or wish to develop, a solid background in machine learning

Project 5.5: Operating Room Body Pose Estimation

- **goal:** estimate the 2D/3D pose of the *surgical team members* from multi-view RGB-D footage of a real operating room
- **dataset:** MVOR
- **your task:** multi-view human pose estimation, use of depth information and camera calibration
- best suitable for students who already have, or wish to develop, a solid background in machine learning

Project 6: deformable blood vessels model

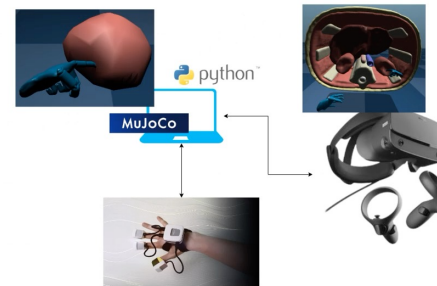


- **goal:** deformable model of blood vessels in the dVRK simulator
- **your task:** model the vessels with standard geometry (e.g., cylinders), train a surrogate 3D model of the vessels and embed in the simulator
- best suitable for students who already have, or wish to develop, a solid background in robot kinematics and machine learning
- what we provide: dVRK in Unity

Project 7, 8, 9, 10: dataset generation

In a nutshell

- VR environment with hand and finger motion tracking
- Haptic feedback with wearable device
- Grasping task of rigid and soft objects
- Dynamic simulation of the interaction with soft tissues (kidney)



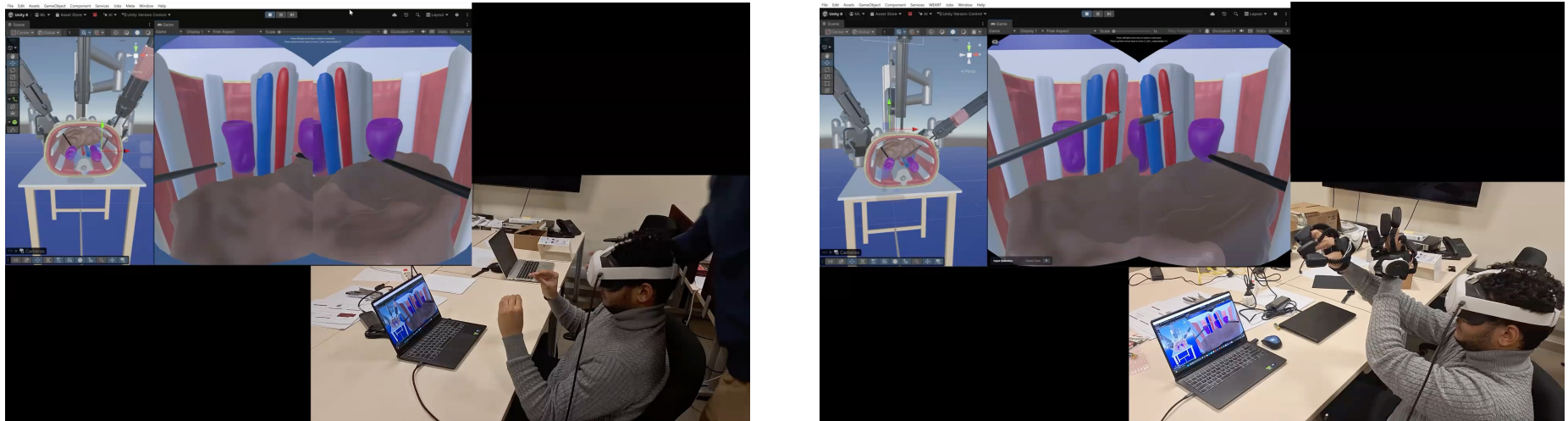
- **goal:** dataset generation for organs of a virtual phantom using [FEBiO](#)
- **your task:** construct the simulation model of different organs of the addominal phantom (liver, kidneys, skin, blood vessels) using realistic physical properties, generate an *interaction dataset* for NN training
- best suitable for students in Biomedical engineering

Project I I: simulation model of the brain



- **goal:** generate a simulation model of the brain to be integrated in Unity or Unreal Engine
- **your task:** construct, or search for already available, 3D model of the brain, import in Unity or Unreal Engine, embed realistic physical properties for simulation
- for all students

Project I2: motion retargeting of hands motion

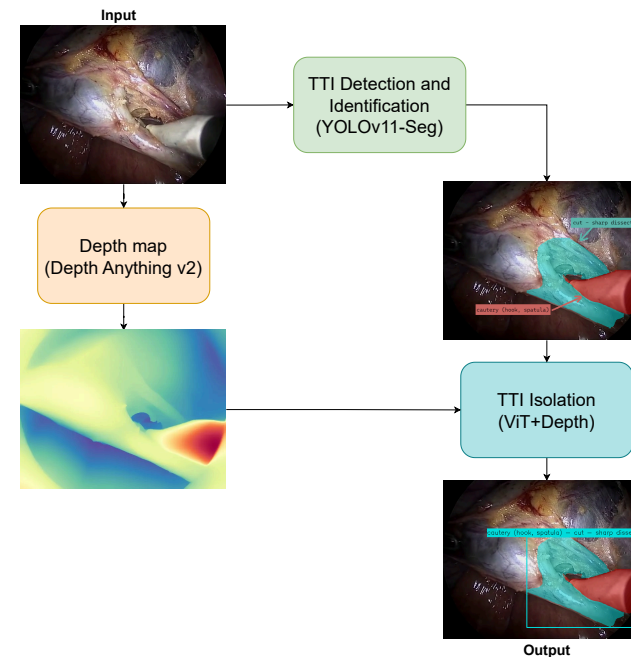
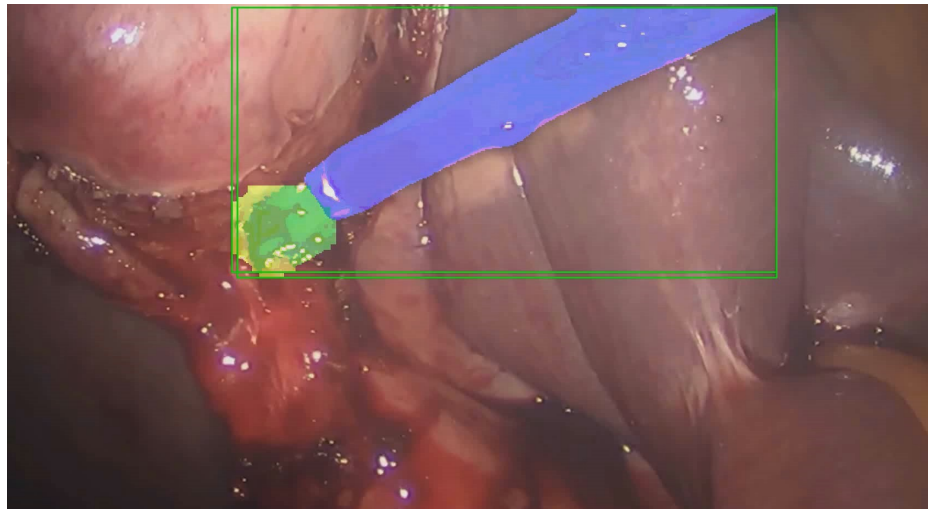


- **goal:** map the motion of human hands on the dVRK simulator tools
- **your task:** determine the most effective retargeting method, add new tools to the simulator, implement in Unity
- for all students

Project 13, 14, 15: soft body simulation

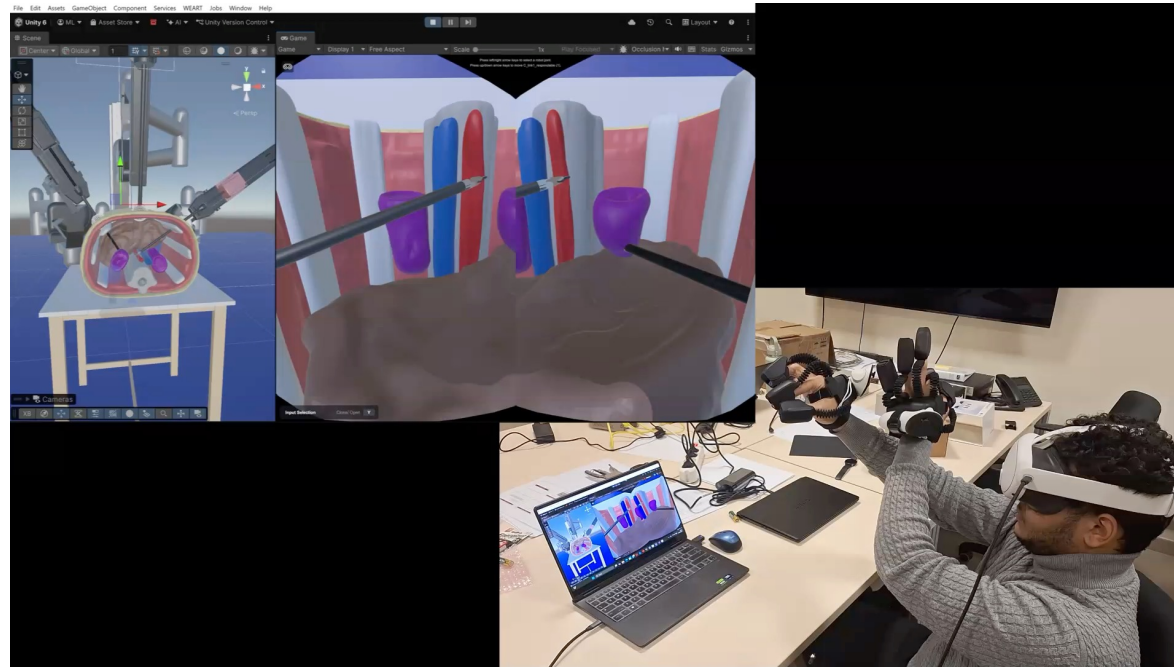
- **goal:** explore different, learning-based, methods for soft body simulation
- causal PINNs
- physics-based DeepONet + (FEM simulation) data
- deformation-force correlation learning: learn in Mujoco, port the model in Unity
- we will provide the bibliographic references
- best suitable for students who already have, or wish to develop, a solid background in machine learning

Project 16: tool-tissue contact detection



- **goal:** explore a different architecture, e.g., U-net+depth anything vs YOLO + depth anything
- we will provide the bibliographic references, dataset and baseline code
- best suitable for students who already have, or wish to develop, a solid background in machine learning

Project I7: optical hand tracking



- **goal:** replace the Weart TouchDiver tracking, requiring to wear the Meta Quest controllers, with optical tracking through the [Polaris camera](#)
- we will provide Polaris camera, localization and tracking algorithms